CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 5: Factorisation

Every formula and result of the chapter on one sheet

1	Common factor	$ab + ac = a(b + c)$
2	Grouping	$ax + ay + bx + by = (a + b)(x + y)$
3	Monic trinomial	$x^2 + (p + q)x + pq = (x + p)(x + q)$
4	General trinomial	Split bx using two numbers with product ac and sum b
5	Difference of squares	$a^2 - b^2 = (a + b)(a - b)$
6	Perfect square	$a^2 + 2ab + b^2 = (a + b)^2$
7	Perfect square	$a^2 - 2ab + b^2 = (a - b)^2$
8	Sum of cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
9	Difference of cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
10	Fourth powers	$a^4 - b^4 = (a^2 + b^2)(a + b)(a - b)$
11	Useful form	$x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$
12	Three cubes	$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$